

# Encoding Nothing at the Edge of Stability: Spiking Reservoir Representations of Affective EEG

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**Abstract**—How representations of affective event-related potentials (ERPs) degrade under perturbation depends on what an evaluation lets stand for nothing: the number replacing a missing electrode, the origin of each feature coordinate, what the pre-stimulus baseline holds after normalization, and the chance floor. We test them on a fixed leaky integrate-and-fire spiking reservoir (six spike-count bins), band-power, ERP-window, and EEGNet, under subject-grouped validation on the SHAPE cohort (211 subjects) and a public IAPS cohort (228 subjects). Moving the coordinate origin leaves clean predictions unchanged, up to solver tolerance, while channel-dropout accuracy falls when the fill moves far from the training mean. The zero-versus-mean fill choice rewrites predictions in the order of feature non-centeredness (34.5% of band-power, 32.9% of ERP-window, 3.6% of reservoir labels) but costs accuracy only as far as the above-chance margin allows. Per-epoch z-scoring writes the post-stimulus mean into the pre-stimulus baseline ( $r = -0.91$ ), enough for pre-stimulus samples alone to classify affect above chance (permutation  $p = 0.025$ ), and the conference reservoir window gains +0.092 BA when moved after onset. Across ten reservoir draws the measured edge of stability lies at  $\rho^* = 0.76 \pm 0.04$ , below  $\rho = 1$ , but the firing threshold moves it from 0.35 to at least 1.57, and past the edge a silent electrode lies 5 to 6 standard deviations from typical reservoir activity. On the long external epoch, spline repair of removed electrodes erases the encoder ranking that zero traces produce. We recommend reporting fills with feature centeredness and above-chance margin, normalization and window placement, and operating points across draws.

**Index Terms**—EEG, event-related potentials, spiking neural networks, reservoir computing, missing data, normalization.

## I. INTRODUCTION

Electroencephalographic event-related potentials (ERPs) measure evoked neural dynamics at millisecond resolution, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [14], [15]. A representation of affective ERPs can separate conditions under one protocol and keep little usable structure after amplitude noise, temporal misalignment, or electrode loss, and endpoint accuracy alone does not say which. Spiking reservoirs make the gap sharper: fixed recurrent networks give nonlinear temporal expansions [1], [4], yet EEG studies

usually score them as classifiers [11], [17] rather than as measurement operators whose invariances can be tested.

Every such test has to decide what *nothing* means. An absent electrode must be represented by some number. A feature coordinate has an origin, and a zero-filled block sits at that origin whether or not it means anything there. The interval before stimulus onset contains no stimulus-locked signal, yet it is part of every epoch the encoder reads. A score has to be compared with the floor set by no information at all. And a reservoir's recurrence can be switched off ( $\rho = 0$ ) to ask what the dynamics add. These choices are rarely stated, and it has not been tested how far each one moves a measured result or whether its effect transfers between cohorts.

Our conference study [30] measured two of them on one cohort. On SHAPE (211 subjects), raw retention rewarded the encoder closest to chance, and zero-filling a dropped channel before standardization, which is not invariant to re-centering the features (Proposition 1), halved the measured lead of ERP-window features over a near-centered spiking-reservoir code at 30% dropout (+0.049 versus +0.092 to +0.107 BA), and on DEAP the fill rule alone moved band-power BA by up to 0.11. The reservoir, a fixed leaky integrate-and-fire (LIF) network read out through six spike-count bins ( $BSC_6$ ), sat just past its measured edge of stability ( $\rho^* \approx 0.82$ , below the echo-state heuristic  $\rho = 1$ ), and removing its recurrence cost 0.049 BA. That study left five questions open. It used one cohort, one reservoir draw, and one EEGNet recipe. It argued Proposition 1 algebraically but never moved the origin to watch the score follow. And its own camera-ready code check found that its reservoir window,  $t \in [10, 70)$  samples, spans about  $-160$  to  $+73$  ms around onset, so most of what the reservoir encoded came before the stimulus.

This paper answers those questions with a second, public cohort and new experiments. The contributions are:

- 1) An external, paradigm-matched test on an independent IAPS cohort (228 subjects, different laboratory and montage) at the conference-matched epoch and at a long epoch pre-specified from the dataset's published effect latencies (Secs. IV, V-A).
- 2) A direct experimental test of Proposition 1: shifting the coordinate origin leaves clean predictions unchanged up to solver tolerance, while the dropout score falls when the fill moves far from the training mean (Sec. V-C).
- 3) Proposition 2, which shows that per-epoch z-scoring writes a function of the post-stimulus signal into the pre-stimulus interval, and a test of how much of each encoder's accuracy that interval carries (Secs. III-G, V-D).

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A preliminary version of this work was accepted at the 2026 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) [30]. This version adds an independent public cohort, a direct test of Proposition 1, Proposition 2 and the pre-stimulus experiments, the operating point across ten reservoir draws, spherical-spline repair and recorded trial-count noise, and EEGNet recipe variation.

- 4) The operating point across ten reservoir draws, thresholds, and leaks: where the measured edge of stability lies and whether accuracy follows it (Sec. V-E).
- 5) Two physically grounded perturbations, spherical-spline repair of removed electrodes and re-averaging from fewer recorded trials, together with EEGNet with and without augmentation and across recipes on the second cohort (Sec. V-G).

The affective categories define the task. The object of study is what each representation does with nothing: with a missing electrode, a shifted origin, an empty baseline, and a chance floor.

## II. RELATED WORK

*ERP representations of affect.* Emotional pictures enlarge the late positive potential relative to neutral ones from roughly 400 ms after onset [43], [46], [47], and a recent cohort of 224 participants adds an ordered valence effect between 1500 and 2500 ms [32]. ERP/EEG decoders commonly use window amplitudes, Hjorth [12] and spectral descriptors [13], and linear models [15], [44], which are interpretable through these known components but can miss nonlinear temporal structure. Deep networks such as EEGNet [16] are now common [17], [45], and much of this literature reports endpoint accuracy over preprocessing and acquisition choices, which can change measured performance materially [18]–[20].

*Reservoirs and the edge of stability.* Reservoir computing maps an input through a fixed high-dimensional recurrent system to a state trajectory read out by a simple classifier [1]–[4], and spiking liquid-state machines add event timing and count structure [7], [8]. Computation is often reported to peak near the transition between ordered and irregular dynamics, located by how far a small state perturbation spreads [5], [6], [34]. The spectral radius  $\rho = 1$  is a heuristic for rate reservoirs, and it is neither necessary nor sufficient for the echo-state property of a driven system [35], so for a thresholded spiking reservoir the transition need not lie at  $\rho = 1$ . Neuromorphic reservoir features for affective EEG were introduced in earlier ARSPI-Net work [9], [10], and this paper extends our conference study [30].

*Missing electrodes.* Dynamic spatial filtering handles missing channels with a learned attention front end [23], and channel-dropout or noise augmentation hardens trained models [24]. A recent EEG foundation-model benchmark finds that apparent channel-dropout fragility depends on whether dropped channels are removed or zero-padded [22]. Fills range from constants and spherical-spline interpolation [25], through learned imputation [23], to generative synthesis of whole missing modalities [26], [27]. The non-learned fills studied here sit at the simple end of that spectrum.

*Evaluation choices that create or remove effects.* Filtering can move late ERP effects to earlier latencies and create artifactual early effects [38]–[40], and leakage between preprocessing and evaluation inflates reported accuracy across machine-learning science [41], [42]. Decoding accuracy near a theoretical chance level needs empirical null distributions [36], and cross-validated accuracy on a few hundred observations

carries wide error bars [37]. Per-epoch normalization, the fill convention for an absent electrode, and the origin of a feature coordinate belong to the same family of choices. We give each a proposition or a direct test.

## III. METHODS

### A. ERP Observation Model

Let  $X_s^{(c)} \in \mathbb{R}^{C \times T}$  denote the trial-averaged ERP observation of subject  $s$  in condition  $c$ . We treat it as a noisy readout of an unobserved neural state trajectory,

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \quad x_t = H\xi_t + \epsilon_t, \quad (1)$$

where  $H$  is the scalp measurement operator and  $\epsilon_t$  collects measurement noise, residual single-trial variability, and pre-processing. Every perturbation below acts on a term of this model: electrode removal zeroes rows of  $H$ , additive noise and re-averaging from fewer trials enlarge  $\epsilon_t$ , and alignment jitter shifts the sampling index of  $x_t$ . Normalization acts on  $x_t$  before any encoder sees it, which is why it is treated as part of the measurement (Sec. III-G).

### B. Spiking Reservoir Encoding

Each electrode trace drives a fixed LIF reservoir of 256 units with input weights  $W_{\text{in}}$ , recurrent weights  $W$ , leak  $\beta = 0.05$ , and threshold  $\vartheta = 0.5$ :

$$v_t = [(1 - \beta)(1 - z_{t-1}) \odot v_{t-1} + Wz_{t-1} + W_{\text{in}}x_t]_+, \quad (2)$$

$$z_t = \mathbf{1}[v_t \geq \vartheta], \quad (3)$$

where  $v_t$  is the membrane potential,  $z_t$  the binary spike vector, and  $[\cdot]_+$  rectification at zero. The factor  $(1 - z_{t-1})$  resets every unit that spiked at the previous step.  $W_{\text{in}}$  and  $W$  are drawn once from a Xavier-uniform law, and  $W$  is rescaled to spectral radius  $\rho$ . Unless stated otherwise the draw is seed 42 and  $\rho = 0.9$ , the conference operating point, and only the readout is trained. Figure 1 shows the reservoir’s activity and the conference operating-point results on SHAPE.

Activity is summarized by six equal temporal bins over an analysis window. For unit  $j$  and bin  $k$ , the binned spike-count code is  $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$ . Each electrode’s 1536-dimensional spike-bin vector is projected onto a single 64-component PCA basis shared across electrodes and fitted on the training subjects of each fold, and the projections are concatenated, giving  $E \in \mathbb{R}^{64C}$  (2176 for SHAPE, 2048 for the external cohort):

$$E = \phi_{\text{PCA}}([b_1, \dots, b_6]) = \Phi(X). \quad (4)$$

A companion study compares this data-fitted projection with a fixed random projection and examines how subject covariance shapes affective EEG embeddings [31]. Here we keep the conference’s train-only PCA so that every result is comparable with the conference version. The conference window is  $t \in [10, 70]$  in samples. With onset at sample 51 in both cohorts it spans about  $-160$  to  $+73$  ms around onset, so most of it precedes the stimulus. Section III-G examines what that window can read, and Sec. V-D adds windows wholly before and wholly after onset.

A silent electrode is a fixed point of the reservoir: with  $x_t = 0$  from a zero initial state,  $v_t$  stays at zero and no unit spikes, so the electrode's spike-bin block is exactly zero. After the PCA step that block maps to  $-P\bar{b}$ , the negative of the projected training mean, not to the origin of the embedding.

### C. Operating Point and Order Parameter

To test whether the recurrent dynamics matter, the same draw is rescaled to  $\rho \in \{0, 0.3, 0.6, 0.9, 1.2, 1.5\}$  with every other parameter fixed ( $\rho = 0$  leaves a feedforward LIF layer). A label-free order parameter locates the dynamical transition by damage spreading [5]: two copies of the reservoir receive the same electrode drive, one unit's spike is flipped at  $t = 10$ , and the normalized Hamming distance between the copies' spike vectors is averaged over steps 21 to 69 and over 400 sampled drives and flipped units. The transition  $\rho^*$  is the  $\rho$  at which this damage, linearly interpolated on the grid  $\rho = 0, 0.1, \dots, 2$ , reaches half its maximum. The conference measured  $\rho^*$  for one draw on SHAPE drives. Here the order parameter is measured for ten independent draws (seeds 42–51) on the external drives of the conference-matched epoch, with the conference definition, and for draw 42 also at thresholds  $\vartheta \in \{0.25, 0.5, 1.0\}$  and leaks  $\beta \in \{0.02, 0.05, 0.1, 0.2\}$ . The accuracy sweep is repeated for five draws (seeds 42–46) on the long epoch of the external cohort (Sec. IV), where the reservoir is above chance.

### D. Baselines

Band-power uses Welch power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz) per channel (5C dimensions). ERP-window uses mean amplitude in three windows, about –120–0 ms, 50–250 ms, and 250–600 ms relative to onset, per channel (3C dimensions). All three fixed encoders feed a balanced L2-regularized logistic readout. As a trained baseline, EEGNet-8,2 [16] is trained per fold on the same per-epoch z-scored  $C \times T$  signal that the fixed encoders read (Adam, learning rate  $10^{-3}$ , 80 epochs, batch 64, dropout 0.25) with train-only per-channel standardization, with and without augmentation by random channel dropout and additive noise. On the external cohort the recipe is also varied (40 or 160 epochs, dropout 0.5) to bound the effect of a single recipe. For cost, the recipe variants and EEGNet on the long epoch use one partition (seed 42, five folds) instead of five, with intervals from the same subject bootstrap.

### E. Perturbations

The feature-coordinate experiment suppresses a dropped channel's feature block after extraction and refills it by one of four rules: zero, train mean, train-fitted  $k$ -nearest-neighbor imputation ( $k=10$ ), and the mean of the retained channels (spatial). It isolates the fill convention and applies to the fixed encoders.

The signal-level experiment corrupts the test subjects' signal before any feature extraction, identically for every encoder including EEGNet, with the same draws in each fold, while training uses the clean signal: removal of 10, 30, or 50% of

the electrodes (a removed electrode is a zero trace), additive white noise at 5 dB SNR per channel, and a uniform alignment shift of up to  $\pm 50$  ms. Two signal-level conditions are new. *Spline repair* replaces each removed electrode by spherical-spline interpolation from the retained electrodes [25] (order 4, seven Legendre terms) on the baseline-corrected microvolt signal before normalization, a physically grounded alternative to the zero trace. *Recorded observation noise* re-averages every test observation from 1, 4, or 16 of its own single trials instead of all of them, so the added variance is the recorded trial-to-trial variability rather than synthetic noise. It needs single trials and is run on the external cohort only.

### F. Retention and the Coordinate Origin

*Raw retention*, the ratio of perturbed to clean BA, overstates the stability of representations whose clean BA is close to chance ( $c = 1/3$ ), so we also report *above-chance retention*,  $(\text{BA}_{\text{pert}} - c)/(\text{BA}_{\text{clean}} - c)$ , and treat absolute BA and paired subject-level differences as primary.

Write the standardized readout input as  $z = D_\sigma^{-1}(x - \mu)$  with  $D_\sigma = \text{diag}(\sigma)$ , and let  $S$  index the dropped coordinates. Zero-fill sets  $x_S = 0$ , so the dropped block enters as  $z_S = -D_{\sigma,S}^{-1}\mu_S$ , and train-mean fill sets  $x_S = \mu_S$ , so  $z_S = 0$ :

$$z_S^{\text{zero}} - z_S^{\text{mean}} = -D_{\sigma,S}^{-1}\mu_S. \quad (5)$$

**Proposition 1 (origin dependence of zero-fill).** *Let the clean readout depend on  $x$  only through  $z$ , so it is invariant to a constant offset  $x \mapsto x - b$  (train-only standardization guarantees this). Then the clean prediction is unchanged by the re-centering, while the zero-fill dropout input shifts by  $\Delta z_S = D_{\sigma,S}^{-1}b_S$ . Hence the measured zero-fill dropout accuracy depends on the coordinate origin, and the zero- and mean-fill inputs coincide if and only if  $\mu_S = 0$ .*

*Proof.* Standardization subtracts the training mean, so  $b$  cancels from the clean input. A zero-filled coordinate maps to  $z_i = -(\mu_i - b_i)/\sigma_i$  instead of  $-\mu_i/\sigma_i$ , a shift of  $b_i/\sigma_i \neq 0$  for any  $b_i \neq 0$ .  $\square$

For a linear readout with class weights  $W$ , Eq. (5) moves the class logits by  $\Delta \ell = -W_S D_{\sigma,S}^{-1}\mu_S$  between the two fills, so the fills can disagree only as far as the dropped features are non-centered. Proposition 1 has a direct experimental test, which the conference study did not run: fill the dropped block with the raw value  $x_i = \mu_i + \kappa\sigma_i$ , which places it at standardized position  $z_i = \kappa$  for every encoder, and sweep  $\kappa$ . Clean predictions must not change with any offset  $b$ , while dropout accuracy must depend on  $\kappa$ , with train-mean fill at  $\kappa = 0$ . An encoder's own zero places each dropped feature at its own  $z_i = -\mu_i/\sigma_i$ , at a mean absolute standardized distance  $|\mu|/\sigma$  from the training mean, so it is not a single point on the  $\kappa$  axis. We refit the readout on features shifted by  $b = \mu + 2\sigma$  to check the first prediction and sweep  $\kappa \in [-3, 3]$  to test the second.

### G. What the Pre-Stimulus Interval Can Contain

Both cohorts are baseline-corrected over the 200 ms before onset, so the pre-stimulus mean of every trial-averaged trace is zero at the recording rate and close to zero after decimation,

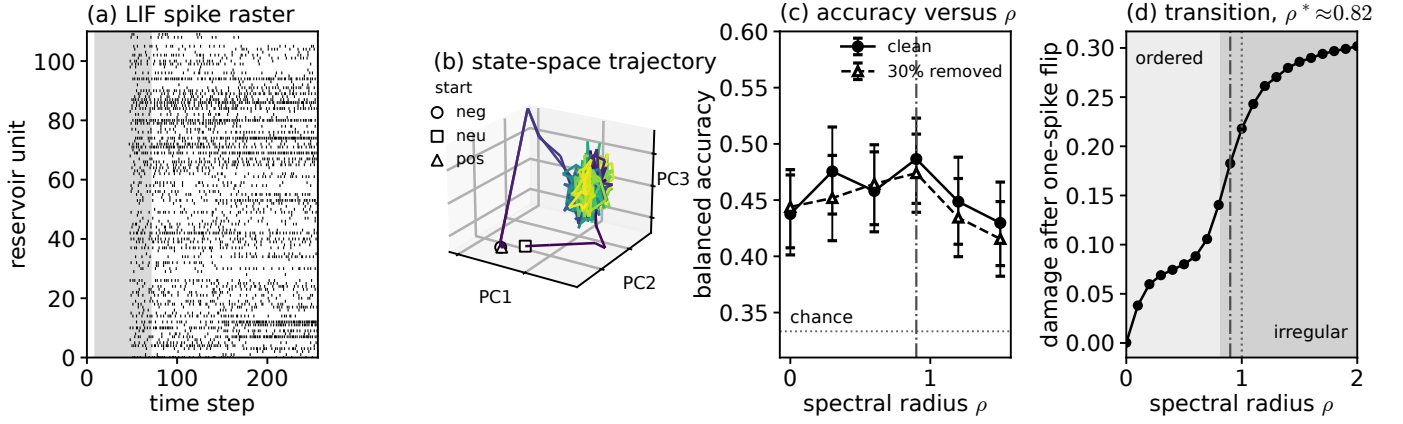


Fig. 1. The fixed LIF reservoir as a dynamical system (SHAPE, draw 42, conference results [30]). (a) Spike raster (units 0–109 of 256) for one condition. Shaded = the conference window  $t \in [10, 70)$ , about  $-160$  to  $+73$  ms around onset at step 51. (b) Population state-space trajectory (first three principal components of the membrane state) for the three conditions. (c) Subject-level BA (95% CI) versus spectral radius  $\rho$ , clean and with 30% of electrodes removed, with dash-dot = the pre-specified  $\rho = 0.9$ . (d) Damage-spreading order parameter versus  $\rho$ . Shading separates the ordered and irregular regimes at  $\rho^* \approx 0.82$ , and dotted = the echo-state heuristic  $\rho = 1$ .

and both are then z-scored per channel over the whole epoch, as in the conference pipeline. Let  $P$  be the pre-stimulus samples and  $Q$  the rest, with  $|P| + |Q| = T$ , and let  $m_Q$  be the post-stimulus mean and  $s$  the epoch standard deviation of one channel.

**Proposition 2 (baseline transfer under per-epoch normalization).** *For a trace with zero pre-stimulus mean normalized by its own epoch mean and standard deviation, the mean of the normalized pre-stimulus segment is  $-\frac{|Q|}{T} m_Q/s$ . The pre-stimulus segment is therefore a deterministic function of the post-stimulus signal, and any encoder that reads only  $P$  has access to  $m_Q/s$ .*

*Proof.* The epoch mean is  $(|P| \cdot 0 + |Q| m_Q)/T$ . Subtracting it and dividing by  $s$  maps the zero pre-stimulus mean to  $-|Q| m_Q/(Ts)$ .  $\square$

With onset at sample 51 of 256,  $|Q|/T = 0.80$ . An affective modulation of the late positive potential changes  $m_Q$ , so under per-epoch z-scoring a classifier can separate conditions from an interval in which no stimulus has yet been shown. A fixed per-channel scale (one standard deviation per channel, pooled over all observations without labels and applied identically to every epoch) leaves the pre-stimulus mean where baseline correction put it and removes this route. Because the scale is shared by all epochs, it cannot carry information from one part of an epoch into another, although it is transductive with respect to subject inclusion. We compare the two normalizations for windows wholly before onset, the conference window, and windows wholly after onset (Sec. V-D).

#### IV. DATA AND PROTOCOL

##### A. SHAPE Cohort

The primary cohort is the trial-averaged affective ERP data of the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project [29], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University: 211 subjects after exclusion of one anomalous record, each contributing negative, neutral, and pleasant observations (633

in total). Each epoch (34 scalp channels at 1024 Hz, 1.2 s, baseline-corrected over the 200 ms pre-stimulus interval) is analyzed as every fourth of its first 1024 samples: 256 samples at 256 Hz spanning  $-200$  to  $+800$  ms, onset at sample 51, z-scored per channel within each epoch (Fig. 2). SHAPE data are held in a restricted environment. Every SHAPE number in this paper is the conference result [30], regenerated from its committed code and aggregate outputs, and no SHAPE analysis was rerun for this version.

##### B. External Cohort

The external cohort is a public IAPS passive-viewing dataset [32], [33] from a different laboratory and montage: 32 scalp electrodes (BioSemi layout) re-referenced to the mastoids and exported at 512 Hz after the dataset authors' topographic interpolation of bad channels, filtering, ICA ocular correction, and artifact rejection. Of 239 participants, the 228 with a complete negative, neutral, and pleasant panel in the preprocessed release are used (684 observations, median 44 trials per observation, range 20 to 54, with high- and low-arousal pictures of one valence pooled). Each observation is built to the SHAPE layout: trial average, baseline correction over  $-200$  to  $0$  ms, every second sample from  $-200$  ms, giving 256 samples at 256 Hz with onset at sample 51, and z-scoring per channel within the epoch. This conference-matched epoch is the primary external analysis. The grand averages reproduce the published effects (Fig. 3): emotional pictures exceed neutral ones by  $2.12 \mu V$  at PO3 between 500 and 1300 ms, and negative pictures exceed pleasant ones by  $1.32 \mu V$  between 1500 and 2500 ms. Most of both effects lies after  $+800$  ms, so we pre-specified, before computing any of its results, a secondary analysis on a long epoch from  $-200$  to  $+2496$  ms (691 samples, same sampling, baseline, and onset). In it the reservoir window covers 0 to  $+2484$  ms after onset (six bins of 106 samples), the ERP-window encoder adds the two published windows (500–1300 and 1500–2500 ms), and band-power and EEGNet see the whole epoch.

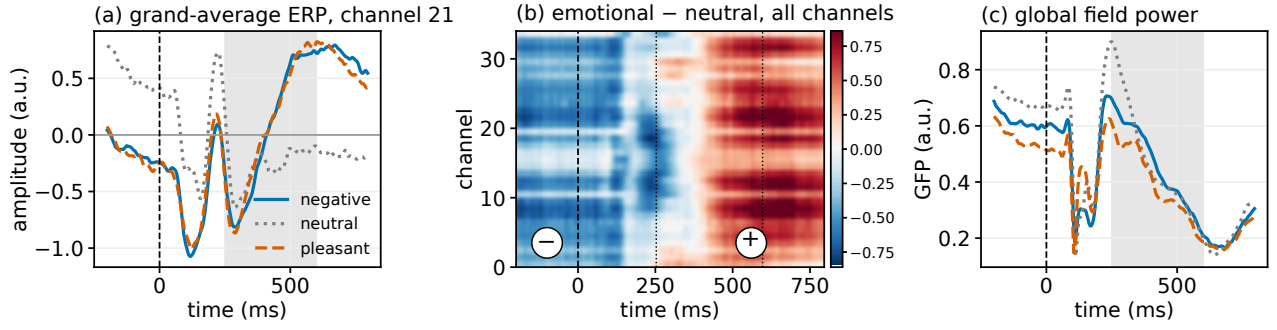


Fig. 2. Trial-averaged affective ERPs of SHAPE (grand averages over 211 subjects, z-scored per epoch, conference results [30]). (a) Waveforms per condition at the most affect-discriminative channel, with time 0 = onset (dashed), shaded = the late ERP-window feature (250–600 ms). (b) Emotional minus neutral across all 34 channels (–/+ mark the sign). (c) Global field power per condition.

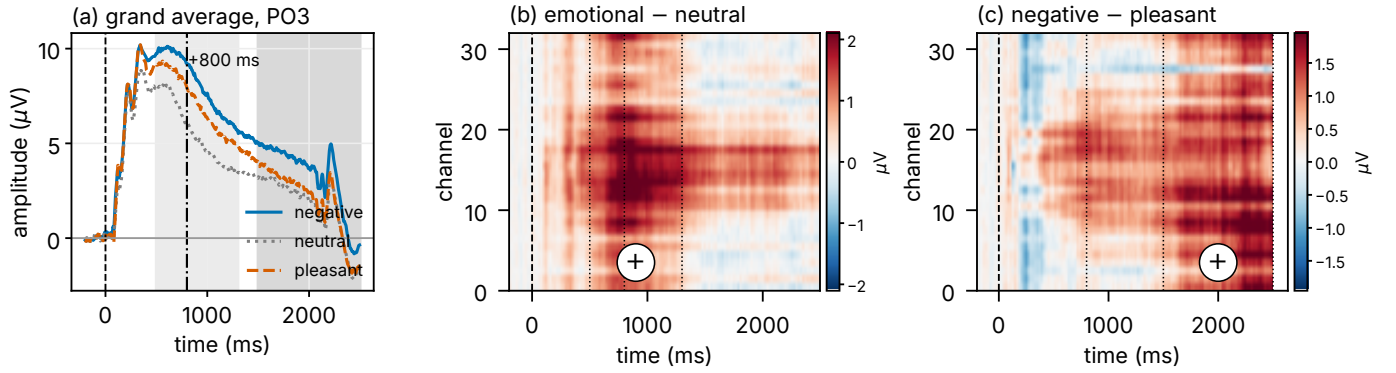


Fig. 3. Grand-average ERPs of the external cohort (228 subjects, baseline-corrected microvolts, long epoch). (a) Waveforms per valence at the channel with the largest emotional-minus-neutral difference between 500 and 1300 ms. Light and dark shading mark the published salience (500–1300 ms) and valence (1500–2500 ms) windows, and the dash-dot line the end of the conference-matched epoch (+800 ms). (b) Emotional minus neutral and (c) negative minus pleasant over all 32 channels, where + marks the sign of the effect in grayscale.

### C. DEAP Check

On DEAP [28] (32 subjects, band-power features, binary valence, within-subject 5-fold) the conference measured how much the fill rule alone moves channel-dropout accuracy. That check is reported unchanged (Fig. 8a).

### D. Validation

All analyses share one fold structure: StratifiedGroupKFold (5 folds, seeds 42–46) with subjects as groups, so a subject’s observations never span training and test. Standardization and the reservoir’s PCA basis are fitted on the training subjects of each fold, the fixed encoders use a balanced L2 logistic readout, and EEGNet is trained on the same folds. The out-of-fold class probabilities of each observation are averaged over the five partitions, and 95% confidence intervals (CIs) come from 4000 resamples of the subjects with replacement, each keeping its three observations. Contrasts between encoders, fills, conditions, or operating points are paired differences under the same resampling, so no test treats the 25 overlapping folds as independent. Chance is 1/3, and a within-subject label-permutation null (20 permutations, in which each subject keeps one observation per class) gives the empirical floor for every fixed encoder [36]. The pre-stimulus test of Sec. V-D uses 200 such permutations and reports  $p = (1 + \#\{\text{null} \geq \text{observed}\}) / (1 + 200)$ . The journal code reproduces the conference implementation bit for bit on synthetic data (reservoir

codes, damage, and bootstrap intervals), and only aggregate results leave the restricted environment.

## V. RESULTS

### A. Clean Classification on Two Cohorts

Table I reports clean three-class BA, macro-F1, macro one-vs-rest ROC-AUC, and the largest BA in the permutation null for every encoder and cohort. On SHAPE, ERP-window features are the strongest fixed encoder (BA 0.629), above band-power (0.494) and the reservoir (0.487), and EEGNet is more accurate still (0.731).

The external cohort carries much less condition information in the same layout. At the conference-matched epoch ERP-window features clearly exceed chance (BA 0.386, CI [0.358, 0.414], against a null maximum of 0.367). Band-power exceeds its null maximum only marginally (0.368 against 0.365, CI [0.339, 0.399]), and the reservoir is at chance (0.320, 95% CI [0.288, 0.354]). On the pre-specified long epoch every fixed encoder clears chance: ERP-window 0.427 [0.398, 0.455], reservoir 0.396 [0.361, 0.428], and band-power 0.389 [0.360, 0.418], and each exceeds its largest permutation BA (null maxima 0.361, 0.387, and 0.377), ERP-window by the widest margin. ERP-window leads the reservoir by +0.030 BA (CI [−0.012, +0.072]). ERP-window is the strongest fixed encoder on both cohorts, although band-power and the reser-

voir swap places on the long epoch, and the external above-chance margin is about a third of SHAPE’s (0.094 versus 0.295 for ERP-window), which bounds every perturbation effect measured on it. The grand averages reproduce the published effects (Fig. 3), so the pipeline recovers the cohort’s ERPs, but the effects that three-class decoding needs are small against the spread between subjects. At PO3, the channel with the largest salience effect (so these effect sizes are optimistic), the salience effect (2.12  $\mu\text{V}$ , within-subject  $d_z = 0.69$ ) is less than half the between-subject standard deviation of the neutral-condition amplitude in the same window (5.58  $\mu\text{V}$ ), and the negative-versus-pleasant effect is weak (1.32  $\mu\text{V}$ ,  $d_z = 0.25$ ) and late.

The reservoir controls of the conference replicate in kind. Pooled (transductive) PCA instead of train-only PCA changes long-epoch BA by  $-0.006$  ( $[-0.018, +0.004]$ ), so the sensitivity check again shows no evidence of optimistic inflation from the transductive basis under this protocol. Component counts of 16, 32, 64, and 128 give 0.428, 0.434, 0.396, and 0.424, and the conference default of 64 gives the lowest of the four, with overlapping intervals. One control changes: over the long window, six bins beat the single-bin total count under 30% electrode removal ( $+0.032$  BA, CI  $[+0.007, +0.056]$ ) and lean the same way on clean data ( $+0.034$ ,  $[-0.010, +0.076]$ ), whereas on SHAPE’s short window BSC<sub>6</sub> and BSC<sub>1</sub> were indistinguishable. This is consistent with a window that spans effects at different latencies, here the salience effect followed by the valence effect, although the clean contrast alone does not exclude zero.

### B. Feature-Coordinate Fills: What Replicates

Proposition 1 names the input-level cause of the conference result: the zero-filled block sits at  $-\mu_S/\sigma_S$  in standardized coordinates. That cause replicates almost exactly. The mean  $|\mu|/\sigma$  of the features is 1.20, 0.75, and 0.06 for band-power, ERP-window, and the reservoir on SHAPE, and 1.20, 0.72, and 0.06 on the external long epoch (1.20, 0.68, 0.05 at the matched epoch). The near-centeredness of the reservoir belongs to its PCA step, not to its spikes: the raw spike-count code is at least as far from centered as band-power (1.14 at the matched epoch, 2.58 on the long window), and the shared basis subtracts the pooled mean.

The accuracy consequence does not replicate at a detectable level (Table II, Fig. 4). On SHAPE, train-mean fill exceeded zero-fill at 30% dropout by  $+0.055$  BA for band-power and  $+0.042$  for ERP-window, and zero-fill halved the ERP-window lead over the reservoir. On the external long epoch the same contrasts are  $-0.013$  ( $[-0.038, +0.012]$ ) and  $+0.014$  ( $[-0.012, +0.039]$ ), and no fill moves the ERP-window lead outside noise. What does replicate is the near-centered code’s stability under the fill choice: the reservoir’s two constant fills differ by at most 0.011 BA at every rate from 10 to 50% on both external epochs, with paired intervals 0.009 to 0.026 BA wide on the long epoch against 0.041 to 0.060 for the non-centered encoders. Section V-C shows why accuracy moved on one cohort and not on the other.

### C. Moving the Origin: A Direct Test of Proposition 1

Proposition 1 makes two predictions that can be tested by moving the coordinate origin. The first holds on both external epochs, and the second on the long epoch, where the encoders have accuracy to lose (Fig. 5, Table III).

First, clean predictions do not depend on the origin. Refitting every readout on features shifted by  $b = \mu + 2\sigma$  changes no predicted label for band-power or the reservoir on either epoch. For ERP-window it changes at most 0.7% of the labels in any fold, near-ties that the stopping tolerance of the L-BFGS solver resolves differently, and class probabilities differ by at most  $5.8 \times 10^{-3}$  (the reservoir’s agree to  $4.2 \times 10^{-14}$ ).

Second, the dropout score depends on where the fill sits. With 30% of the electrodes dropped on the long epoch, ERP-window BA is 0.406 with the fill at the training mean ( $\kappa = 0$ ) and 0.345 with the same block placed three standard deviations away ( $\kappa = +3$ ), a paired loss of 0.062 (CI  $[+0.034, +0.089]$ ), and ERP-window also loses at  $\kappa = \pm 2$  (CIs exclude zero). Band-power falls from 0.385 to 0.349 (a loss of 0.036, CI  $[+0.001, +0.069]$ ). The reservoir is not exempt. Its zero sits about 0.06 standard deviations from its training mean, so the zero-versus-mean choice is immaterial for it, but at  $\kappa = +3$  it loses 0.041 (CI  $[+0.007, +0.073]$ ), comparable to band-power and less than ERP-window. For band-power and the reservoir only the  $\kappa = +3$  contrast excludes zero, and at the conference-matched epoch, where the encoders are near chance, no  $\kappa$  contrast does. Each encoder’s native zero places its dropped features at a mean absolute standardized distance of 1.20 (band-power), 0.72 (ERP-window), and 0.06 (reservoir) from the training mean.

The mechanism replicates even where accuracy cannot move. Between zero-fill and mean-fill, the class logits of a test observation move by  $\|\Delta\ell\| = 1.76$  for band-power, 1.45 for ERP-window, and 0.17 for the reservoir (class-centered norm, mean over folds), and the fill choice changes the predicted label of 34.5%, 32.9%, and 3.6% of the test observations, in the order of the encoders’ non-centeredness. At the conference-matched epoch, where no encoder is far from chance, the same ordering holds (37.8%, 25.9%, and 3.3% of labels change) while BA barely moves. On SHAPE the fill choice cost band-power 0.055 BA and halved an encoder difference (the conference stored no label-change rates). On the external cohort it rewrites about a third of the non-centered encoders’ labels yet costs little accuracy, because the encoders have about a third of SHAPE’s above-chance margin to lose (0.32 for ERP-window, 0.34 for band-power). Proposition 1 thus predicts which representations a fill rule perturbs, and the above-chance margin sets how much accuracy that perturbation can remove.

### D. The Empty Baseline

Proposition 2 predicts that under per-epoch z-scoring the pre-stimulus mean of each normalized trace is a scaled negative of its post-stimulus mean. Across the  $684 \times 32$  external traces the correlation is  $-0.90$  at the conference-matched epoch and  $-0.91$  on the long epoch, so the transfer happens. Whether it carries condition information depends on

TABLE I

CLEAN THREE-CLASS BALANCED ACCURACY (SUBJECT-LEVEL BOOTSTRAP 95% CI), MACRO-F1, MACRO ONE-VS-REST ROC-AUC FROM POOLED OUT-OF-FOLD PROBABILITIES, AND THE LARGEST BA AMONG 20 WITHIN-SUBJECT LABEL PERMUTATIONS. SHAPE VALUES ARE THE CONFERENCE RESULTS [30]. THE CONFERENCE STORED NO EEGNET PROBABILITIES, SO F1 AND AUC ARE NOT AVAILABLE FOR IT. CHANCE IS 1/3 (AUC 0.5).

| Cohort                   | Encoder       | BA [95% CI]          | F1    | AUC   | Null max |
|--------------------------|---------------|----------------------|-------|-------|----------|
| SHAPE ( $N=211$ )        | ERP-window    | 0.629 [0.597, 0.660] | 0.629 | 0.799 | 0.390    |
|                          | Band-power    | 0.494 [0.460, 0.528] | 0.492 | 0.682 | 0.378    |
|                          | Reservoir     | 0.487 [0.447, 0.523] | 0.485 | 0.658 | 0.379    |
|                          | EEGNet        | 0.731 [0.701, 0.763] | –     | –     | –        |
|                          | EEGNet + aug. | 0.731 [0.700, 0.763] | –     | –     | –        |
| External, $-200..800$ ms | ERP-window    | 0.386 [0.358, 0.414] | 0.385 | 0.561 | 0.367    |
|                          | Band-power    | 0.368 [0.339, 0.399] | 0.367 | 0.531 | 0.365    |
|                          | Reservoir     | 0.320 [0.288, 0.354] | 0.320 | 0.482 | 0.368    |
|                          | EEGNet        | 0.501 [0.471, 0.534] | 0.501 | 0.666 | –        |
|                          | EEGNet + aug. | 0.523 [0.491, 0.556] | 0.522 | 0.679 | –        |
| External, long epoch     | ERP-window    | 0.427 [0.398, 0.455] | 0.427 | 0.597 | 0.361    |
|                          | Band-power    | 0.389 [0.360, 0.418] | 0.388 | 0.553 | 0.377    |
|                          | Reservoir     | 0.396 [0.361, 0.428] | 0.396 | 0.576 | 0.387    |
|                          | EEGNet        | 0.447 [0.415, 0.478] | 0.447 | 0.635 | –        |
|                          | EEGNet + aug. | 0.488 [0.455, 0.523] | 0.488 | 0.672 | –        |

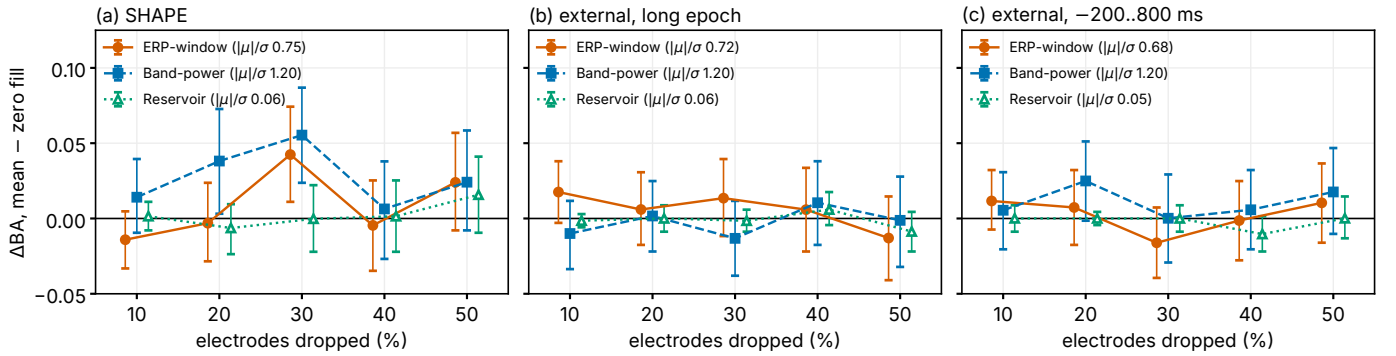


Fig. 4. How much the fill rule alone moves channel-dropout accuracy: paired subject-level change in BA (95% CI) of train-mean fill minus zero-fill against the share of electrodes dropped, for each fixed encoder. The legend gives the measured mean  $|\mu|/\sigma$  of each encoder’s features on that cohort. (a) SHAPE (conference results). (b) External cohort, long epoch. (c) External cohort, conference-matched epoch. The input-level cause ( $|\mu|/\sigma$ ) replicates on both cohorts, but the accuracy change does not, because the external encoders have little accuracy above chance to lose.

how much the post-stimulus mean differs between conditions (Fig. 6). ERP-window features computed only from the 200 ms before onset reach BA 0.357 [0.333, 0.382] under per-epoch z-scoring at the matched epoch (permutation  $p = 0.075$ ) and 0.325 under a fixed per-channel scale. On the long epoch, whose post-stimulus mean now contains the salience and valence effects, the same pre-stimulus features reach 0.367 [0.342, 0.392] under z-scoring (permutation  $p = 0.025$ ) and 0.325 [0.291, 0.358] under the fixed scale (paired difference +0.042, CI [−0.001, +0.085]). A linear readout thus classifies affect above chance from samples recorded before the picture appeared when the epoch is z-scored, and not when it is scaled by a fixed constant, although the paired difference between the two normalizations does not exclude zero. The same normalization removes information where it belongs: on the long epoch, post-onset ERP-window features reach 0.447 under the fixed scale and 0.406 under z-scoring (paired −0.041, CI [−0.075, −0.007]), because dividing each epoch by its own standard deviation discards the amplitude differences that carry the effect.

Under the conference normalization, the window decides what the reservoir measures. With the conference window (−160 to +73 ms) the reservoir is at chance at the matched epoch (0.320, [0.288, 0.354]). Moved wholly after onset (0 to +797 ms, same reservoir, same six bins), it reaches 0.412 [0.377, 0.449], a gain of +0.092 BA (CI [+0.045, +0.139]), and its point estimate then exceeds that of the ERP-window encoder (0.386). A window wholly before onset gives 0.311. Under the fixed scale the matched-epoch gain is +0.045 (CI [−0.006, +0.094]), and on the long epoch the post-onset gain is +0.046 under z-scoring (CI [+0.000, +0.095]) and +0.056 (CI [+0.012, +0.101]) under the fixed scale. For SHAPE, where the conference reservoir reached 0.487 from the conference window, these results cannot divide that accuracy between the 73 ms after onset and the transferred offset, but they show that the reservoir measures the evoked signal when it is pointed at it.

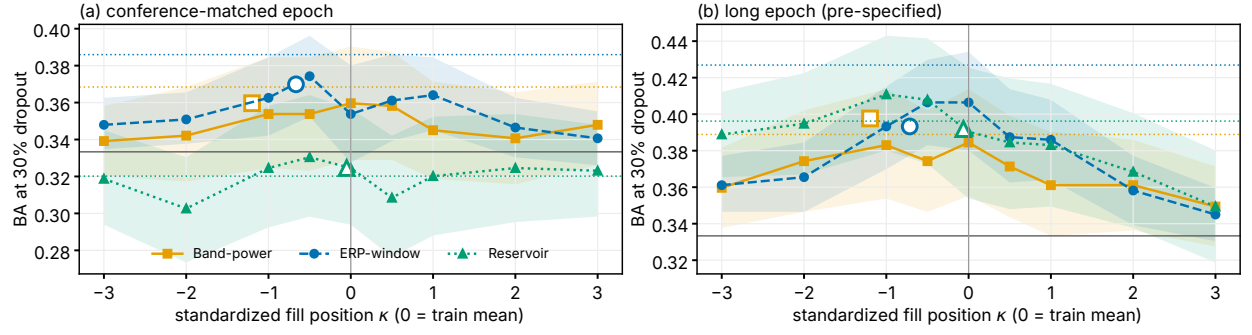


Fig. 5. Moving the fill along the standardized axis (external cohort, 30% of electrodes dropped). Subject-level BA (95% CI, shaded) versus the standardized position  $\kappa$  of the filled block ( $\kappa = 0$ : train-mean fill). Open markers: each encoder’s native zero-fill, plotted at minus the mean  $|\mu|/\sigma$  of its dropped block for reference (the native zero places each feature at its own  $-\mu_i/\sigma_i$ , so it is not a single point on the  $\kappa$  axis). Dotted lines: clean BA, identical for every origin. (a) Conference-matched epoch, where every encoder is near chance. (b) Long epoch: moving the fill to  $\kappa = +3$  costs every encoder, including the reservoir, and ERP-window loses at every  $|\kappa| \geq 2$ .

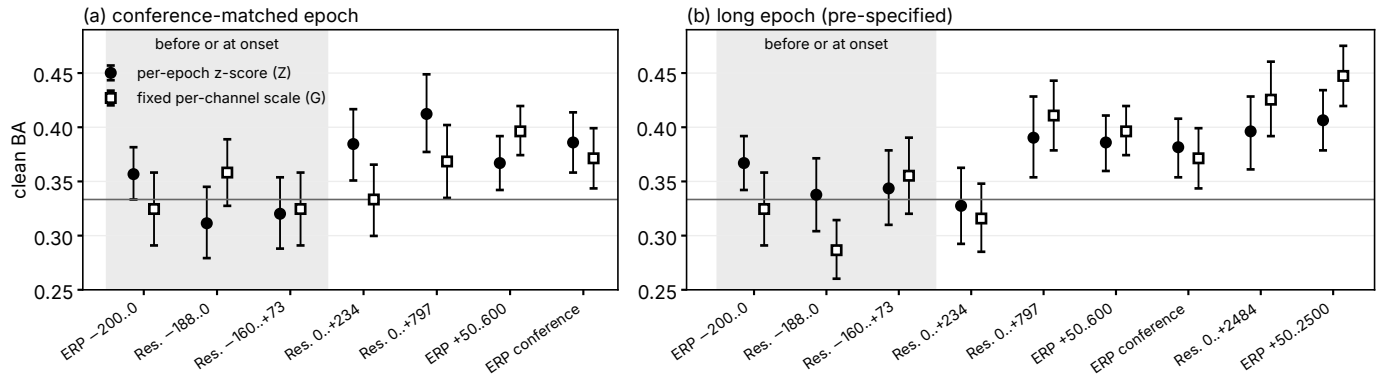


Fig. 6. What each window can read (external cohort). Subject-level BA (95% CI) under per-epoch z-scoring (filled) and a fixed per-channel scale (open), for windows before or at onset (shaded) and after onset. (a) Conference-matched epoch. (b) Long epoch, with two added windows after onset (0 to +2484 ms for the reservoir, the four post-onset ERP windows). Under z-scoring, pre-stimulus ERP-window features exceed chance on the long epoch. Chance =  $1/3$ .

TABLE II

CHANNEL DROPOUT AT 30% (FEATURE-COORDINATE EXPERIMENT): SUBJECT-LEVEL BA UNDER FOUR FILL RULES, AND THE PAIRED DIFFERENCE ERP-WINDOW MINUS RESERVOIR UNDER THE SAME SUBJECT RESAMPLING. EXT. LONG AND EXT. MATCHED: EXTERNAL COHORT, LONG AND CONFERENCE-MATCHED EPOCHS.

| Cohort       | Fill    | Band  | ERP   | Res.  | ERP-Res [95% CI]        |
|--------------|---------|-------|-------|-------|-------------------------|
| SHAPE        | Zero    | 0.417 | 0.536 | 0.487 | +0.049 [+0.003, +0.095] |
|              | Mean    | 0.472 | 0.578 | 0.487 | +0.092 [+0.049, +0.134] |
|              | kNN     | 0.483 | 0.610 | 0.507 | +0.103 [+0.062, +0.147] |
|              | Spatial | 0.488 | 0.589 | 0.482 | +0.107 [+0.063, +0.150] |
| Ext. long    | Zero    | 0.398 | 0.393 | 0.392 | +0.001 [−0.039, +0.042] |
|              | Mean    | 0.385 | 0.406 | 0.390 | +0.017 [−0.025, +0.059] |
|              | kNN     | 0.380 | 0.412 | 0.396 | +0.016 [−0.026, +0.060] |
|              | Spatial | 0.385 | 0.398 | 0.404 | −0.006 [−0.048, +0.035] |
| Ext. matched | Zero    | 0.360 | 0.370 | 0.325 | +0.046 [+0.006, +0.085] |
|              | Mean    | 0.360 | 0.354 | 0.325 | +0.030 [−0.012, +0.072] |
|              | kNN     | 0.376 | 0.389 | 0.319 | +0.070 [+0.025, +0.113] |
|              | Spatial | 0.377 | 0.383 | 0.320 | +0.063 [+0.022, +0.104] |

TABLE III

ORIGIN TEST AT 30% DROPOUT (EXTERNAL COHORT, LONG EPOCH): MEAN  $|\mu|/\sigma$  OF THE DROPPED BLOCK, CLEAN BA, BA WITH THE NATIVE ZERO-FILL AND WITH THE FILL AT  $\kappa = -2, 0, +2$ , THE CLASS-CENTERED LOGIT SHIFT BETWEEN ZERO AND MEAN FILL, AND THE PERCENTAGE OF TEST LABELS THAT THE ZERO-VERSUS-MEAN CHOICE CHANGES.

| Encoder    | $\frac{ \mu }{\sigma}$ | Clean | Zero  | $\kappa = -2$ | $\kappa = 0$ | $\kappa = +2$ | $\ \Delta\ell\ $ | Flip % |
|------------|------------------------|-------|-------|---------------|--------------|---------------|------------------|--------|
| Band-power | 1.20                   | 0.389 | 0.398 | 0.374         | 0.385        | 0.361         | 1.76             | 34.5   |
| ERP-window | 0.72                   | 0.427 | 0.393 | 0.365         | 0.406        | 0.358         | 1.45             | 32.9   |
| Reservoir  | 0.06                   | 0.396 | 0.392 | 0.395         | 0.390        | 0.368         | 0.17             | 3.6    |

### E. The Edge of Stability Across Reservoir Draws

The conference located the transition of draw 42 at  $\rho^* \approx 0.82$  on SHAPE drives. On the external drives the same draw gives  $\rho^* = 0.82$ , and its order-parameter curve agrees with the SHAPE curve within 0.004 at every grid point from  $\rho = 0$  to

2 (the ten-draw mean in Fig. 7a saturates lower), consistent with a transition set by the reservoir and the statistics of its z-scored drive rather than by the cohort. Across ten draws,  $\rho^* = 0.76 \pm 0.04$  (range 0.69 to 0.84, Table IV). Every draw places the transition below the echo-state heuristic  $\rho = 1$ , and the conference draw lies near the top of the range. The unit moves the transition more than the draw does. At thresholds 0.25, 0.5, and 1.0 the transition lies at 0.35, 0.82, and 1.57 (Fig. 7c, where at  $\vartheta = 1.0$  the damage has not saturated by  $\rho = 2$ , so this value is a lower bound), and from leak 0.02 to 0.2 it rises from 0.79 to 0.95. The heuristic  $\rho = 1$  therefore marks no transition for this spiking unit: depending on the threshold, the measured edge lies well below it or above it.

Accuracy on the long epoch follows the conference pattern in shape but not in strength (Fig. 7b, Table IV). Averaged over five draws, clean BA is lowest without recurrence ( $\rho = 0$ , 0.372) and in the strongly irregular regime ( $\rho = 1.5$ , 0.373), and highest at  $\rho = 1.2$  (0.413), the most accurate setting in four of five draws, against 0.385 at the conference operating point. No single draw, however, separates  $\rho = 0$  or  $\rho = 1.5$  from  $\rho = 0.9$  with a subject-level interval that excludes zero, so the recurrence effect measured on SHAPE ( $-0.049$  BA, CI  $[-0.098, -0.002]$ ) is not established on the external cohort, whose margin is smaller. The optimum is broad, spans the measured transition, and does not track  $\rho^*$  draw by draw.

Electrode removal separates the regimes more sharply than clean accuracy does. With 30% of the electrodes silent, mean BA over the five draws is 0.386 and 0.389 at  $\rho = 0.3$  and 0.6 but 0.339, 0.339, and 0.340 at  $\rho = 0.9$ , 1.2, and 1.5, close to chance on average (single draws stay above chance in some cases). The loss to removal is largest past the transition (0.012 BA at  $\rho = 0.6$ , 0.045 at 0.9, 0.074 at 1.2, and 0.033 at 1.5), and so does the firing rate (0.05, 0.14, and 0.23 spikes per unit and step). A silent electrode is a fixed point of every reservoir, so its block is zero spikes whatever  $\rho$  is. The more active the reservoir, the farther that silence lies from its typical state. For draw 42, in standardized readout coordinates the silent block sits on average 1.21 to 1.36 standard deviations from the training mean for  $\rho \leq 0.6$ , but 4.95 at  $\rho = 0.9$  and 6.23 at  $\rho = 1.2$ , and the raw spike counts move from  $|\mu|/\sigma = 0.90$  to 4.40. Past the measured edge of stability, nothing, in the literal sense of a silent electrode, lies far from anything the readout has seen. SHAPE, read through the short conference window, showed no such separation: the loss to 30% removal was 0.013, 0.014, and 0.014 BA at  $\rho = 0.9$ , 1.2, and 1.5.

#### F. DEAP Check

On DEAP [28] (conference result, Fig. 8a), band-power features scored 0.540 under zero-fill at 30% dropout but 0.621 under train-mean and 0.638 under  $k$ NN fill, against 0.639 clean, so the fill rule alone moved dropout accuracy by up to 0.11 BA on a second dataset, task, and montage. Band-power had a margin of 0.14 above the binary chance level there, enough for the fill-induced label changes to cost accuracy, as on SHAPE.

#### G. Signal-Level Perturbations, Physical Repair, and Recorded Noise

At the signal level every encoder sees the same corrupted test signal (Tables V and VI, Fig. 9). On SHAPE (conference results), ERP-window features led the reservoir in every condition, the reservoir held its accuracy through 30% electrode removal but fell to 0.393 at 50%, 5 dB noise cost band-power 0.059 BA, and  $\pm 50$  ms jitter moved no fixed encoder.

*The chance floor in practice.* At the external conference-matched epoch, where no fixed encoder is far from chance, perturbations move BA in both directions: the reservoir scores 0.320 clean and 0.370 under 5 dB noise, a raw retention of 1.16. A ratio above one here measures nothing about stability. It is the raw-retention artifact of the conference paper

appearing without being sought. We therefore read the external signal-level results on the long epoch.

*Electrode removal and spline repair.* On the long epoch, removing electrodes as zero traces costs the reservoir 0.038 BA at 30% (CI  $[+0.004, +0.073]$ ) and 0.051 at 50% ( $[+0.015, +0.088]$ ), and band-power 0.043 at 50%, while ERP-window features lose 0.011 at 50%. ERP-window therefore leads the reservoir by +0.070 (CI  $[+0.041, +0.101]$ ) with half the electrodes removed, a larger lead than on clean data. Repairing the same electrodes by spherical-spline interpolation (correlation 0.84 with the true trace at 50%) removes that lead: with the repaired signal no fixed encoder loses more than 0.011 BA at 50%, the reservoir gains +0.062 over the zero trace (CI  $[+0.025, +0.101]$ ), band-power gains +0.050 ( $[+0.018, +0.082]$ ), and the ERP-window lead falls to +0.010 ( $[-0.032, +0.051]$ ). The signal-level ranking under electrode loss was set by representing the absent electrode as a zero trace. A zero trace is a silent drive, and for the reservoir silence lies far from its typical state (Sec. V-E).

*Noise and jitter.* The SHAPE pattern replicates: 5 dB noise costs band-power 0.047 BA (CI  $[+0.015, +0.079]$ , against 0.059 on SHAPE) and leaves ERP-window and the reservoir within noise, and  $\pm 50$  ms jitter moves no fixed encoder.

*Recorded observation noise.* Re-averaging each test observation from fewer of its own trials lowers ERP-window BA from 0.427 to 0.367 at one trial (loss 0.060, CI  $[+0.018, +0.102]$ ), whereas the reservoir is unchanged at one trial (0.396, loss 0.000, CI  $[-0.045, +0.045]$ ) and band-power loses 0.023 ( $[-0.019, +0.064]$ ) (Fig. 9a). The paired ERP-window minus reservoir difference moves from +0.030 on clean data to  $-0.029$  at one trial (CI  $[-0.072, +0.015]$ ), so the encoders are not separable on single trials. Training used the full averages throughout, so these numbers measure how each fixed representation transfers from trial averages to noisier observations.

*Training effect.* EEGNet, trained end to end on the signal, is the most accurate encoder on the external cohort as on SHAPE. At the conference-matched epoch, where every fixed encoder is near chance, it reaches 0.501 (CI  $[0.471, 0.534]$ ) without and 0.523 with augmentation, +0.116 above ERP-window (CI  $[+0.076, +0.154]$ ), and its paired lead over ERP-window has a lower bound at or above zero in every condition (Table VII). The SHAPE training effect does not replicate. On SHAPE, channel-dropout and noise augmentation cut EEGNet's loss to electrode removal by more than half (0.144 to 0.062 BA at 50%). Here the loss is 0.058 and 0.050 at 30 and 50% without augmentation and 0.051 and 0.057 with it. Augmentation instead adds at most +0.041 BA in any condition, with intervals that exclude zero only under 10% removal, 30% spline repair, and 5 dB noise ( $+0.041$ , CI  $[+0.010, +0.072]$ ). Spline repair recovers most of the removal loss for EEGNet as for the fixed encoders (loss 0.009 at 50%, CI  $[-0.018, +0.037]$ ). The trained network is also the encoder most affected by recorded observation noise: re-averaging test observations from one trial costs it 0.131 BA (CI  $[+0.092, +0.171]$ ), against 0.055 for ERP-window, consistent with a network fitted to trial averages. On the long epoch (seed-42 partition), EEGNet reaches 0.447 without and 0.488 with augmentation, against 0.463 and 0.485

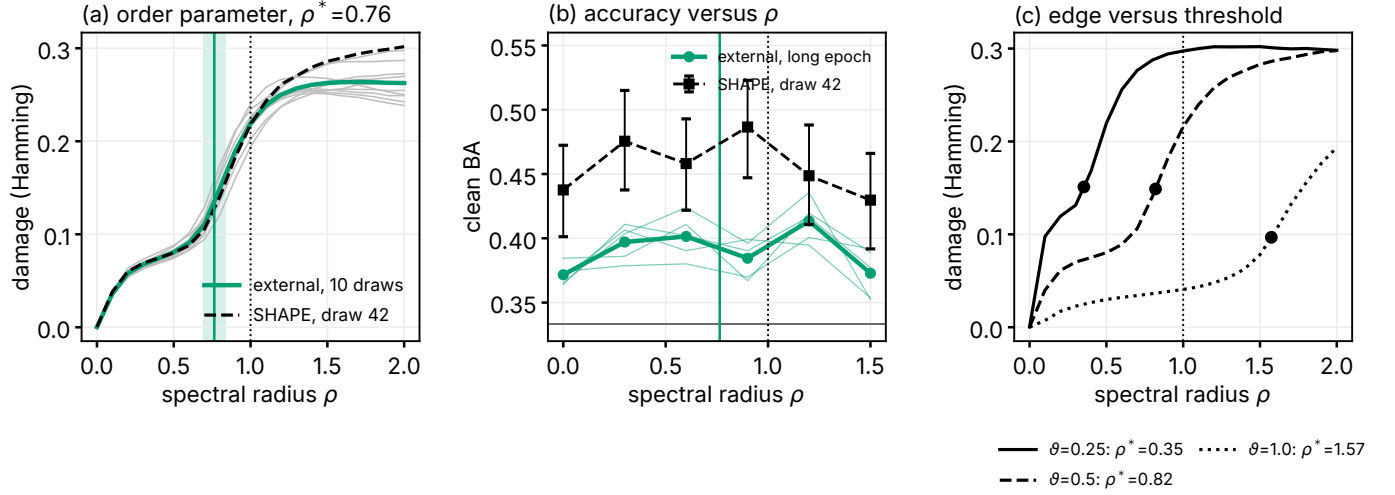


Fig. 7. The measured edge of stability. (a) Damage-spreading order parameter versus spectral radius for ten reservoir draws on the external drives (thin), their mean (thick), and SHAPE (draw 42, dashed). Shading spans the transitions  $\rho^*$  of the ten draws, the solid line marks their mean, and the dotted line the echo-state heuristic  $\rho = 1$ . (b) Clean BA versus  $\rho$  for five draws on the external long epoch (thin) and their mean (thick), with SHAPE (draw 42, 95% CI).  $\rho = 0$  removes the recurrence, and chance =  $1/3$ . (c) Order parameter at three firing thresholds (draw 42), with dots at each  $\rho^*$ .

TABLE IV  
EDGE OF STABILITY ACROSS RESERVOIR DRAWS (EXTERNAL COHORT): MEASURED TRANSITION  $\rho^*$  (ORDER PARAMETER ON THE CONFERENCE-MATCHED DRIVES) AND CLEAN BA ON THE LONG EPOCH AT EACH SPECTRAL RADIUS, THE MOST ACCURATE  $\rho$ , AND THE PAIRED DIFFERENCE BETWEEN  $\rho = 0$  AND  $\rho = 0.9$  UNDER THE SAME SUBJECT RESAMPLING.

| Draw | $\rho^*$ | 0     | 0.3   | 0.6   | 0.9   | 1.2   | 1.5   | Best | $\rho=0 - 0.9$ [95% CI] |
|------|----------|-------|-------|-------|-------|-------|-------|------|-------------------------|
| 42   | 0.82     | 0.365 | 0.404 | 0.424 | 0.396 | 0.436 | 0.352 | 1.2  | -0.031 [-0.077, +0.016] |
| 43   | 0.69     | 0.364 | 0.411 | 0.402 | 0.390 | 0.417 | 0.377 | 1.2  | -0.026 [-0.073, +0.019] |
| 44   | 0.76     | 0.374 | 0.379 | 0.380 | 0.370 | 0.401 | 0.392 | 1.2  | +0.004 [-0.047, +0.054] |
| 45   | 0.76     | 0.370 | 0.406 | 0.390 | 0.399 | 0.395 | 0.354 | 0.3  | -0.029 [-0.072, +0.016] |
| 46   | 0.78     | 0.385 | 0.386 | 0.411 | 0.367 | 0.420 | 0.389 | 1.2  | +0.017 [-0.029, +0.064] |
| Mean | 0.76     | 0.372 | 0.397 | 0.401 | 0.385 | 0.413 | 0.373 |      |                         |

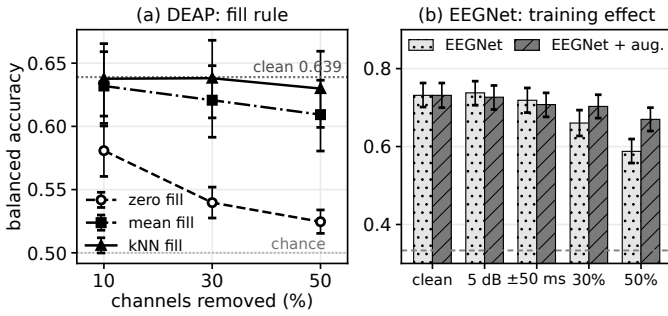


Fig. 8. Conference results [30]. (a) DEAP [28] (32 subjects, within-subject binary valence, band-power, CIs from resampling subjects): under channel removal, BA depends strongly on the fill rule. (b) SHAPE, EEGNet without and with channel-dropout and noise augmentation under the signal-level conditions of Table V (subject-level 95% CIs): augmentation leaves clean accuracy unchanged and cuts the loss to electrode removal by more than half.

at the matched epoch on the same partition, so with this recipe the late effects that help the fixed encoders do not raise its accuracy. Augmentation again adds accuracy across conditions (+0.041 on clean data, CI [+0.010, +0.073]) without reducing the loss to 50% removal (0.053 without and 0.064 with augmentation), and spline repair again recovers most of that loss (0.013).

TABLE V  
SHAPE, SIGNAL-LEVEL PERTURBATIONS (CONFERENCE RESULTS [30]): SUBJECT-LEVEL BA WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED TEST SIGNAL. PER-CELL 95% CIs ARE  $\pm 0.03$  TO  $\pm 0.04$ .

| Condition          | Band  | ERP   | Res.  | EEGNet | +aug. |
|--------------------|-------|-------|-------|--------|-------|
| Clean              | 0.494 | 0.629 | 0.487 | 0.731  | 0.731 |
| 10% removed        | 0.476 | 0.602 | 0.469 | 0.719  | 0.722 |
| 30% removed        | 0.417 | 0.536 | 0.474 | 0.660  | 0.703 |
| 50% removed        | 0.450 | 0.474 | 0.393 | 0.588  | 0.670 |
| Noise 5 dB         | 0.436 | 0.619 | 0.479 | 0.738  | 0.727 |
| Jitter $\pm 50$ ms | 0.496 | 0.626 | 0.493 | 0.719  | 0.708 |

## VI. DISCUSSION

*Absence is a coordinate.* The conference result was that zero-fill halves an encoder difference. The two-cohort result is more specific. What transfers between cohorts is the input-level geometry. Each encoder's dropped block sits at nearly the same standardized distance from its training mean on both cohorts, on the external cohort the fill choice rewrites predictions in the order of that distance, and moving the fill along the standardized axis changes the dropout score while clean predictions stay fixed up to solver tolerance. What does not transfer is the size of the accuracy change, because a fill can only cost what an encoder has above chance. A dropout

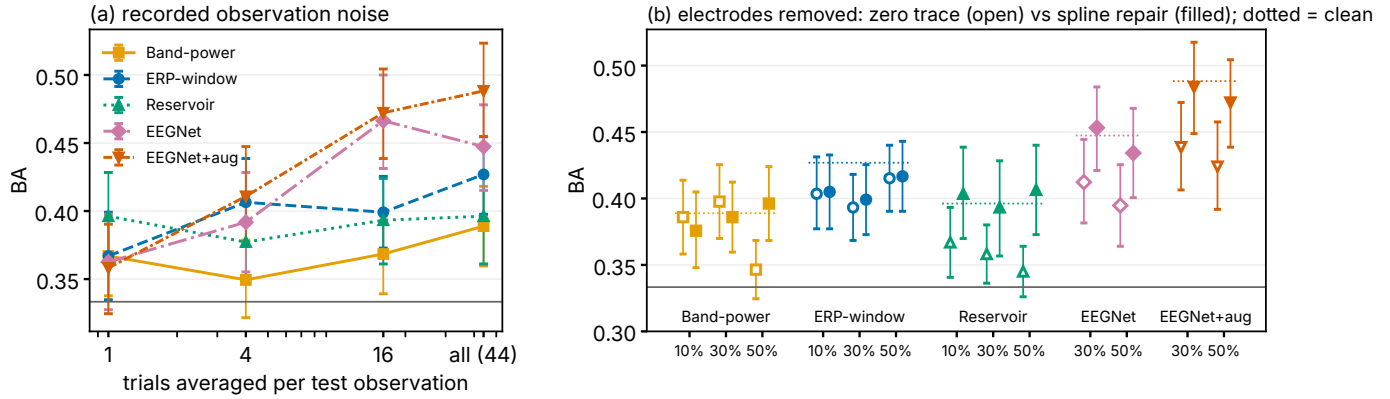


Fig. 9. Signal-level perturbations on the external cohort (long epoch). (a) Recorded observation noise: subject-level BA (95% CI) when each test observation is re-averaged from 1, 4, or 16 of its single trials, and with all trials. (b) Electrodes removed as a zero trace (open) or repaired by spherical-spline interpolation (filled), with dotted lines at clean BA. Chance = 1/3.

TABLE VI

EXTERNAL COHORT, LONG EPOCH: SUBJECT-LEVEL BA WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED TEST SIGNAL (SAME DRAWS FOR ALL ENCODERS). SPLINE: THE REMOVED ELECTRODES REPAIRED BY SPHERICAL-SPLINE INTERPOLATION.  $n$  TRIALS: EACH TEST OBSERVATION RE-AVERAGED FROM  $n$  OF ITS OWN TRIALS. EEGNET ON THIS EPOCH USES THE SEED-42 PARTITION ONLY. PER-CELL 95% CIS ARE  $\pm 0.02$  TO  $\pm 0.04$ .

| Condition          | Band  | ERP   | Res.  | EEGNet | +aug. |
|--------------------|-------|-------|-------|--------|-------|
| Clean              | 0.389 | 0.427 | 0.396 | 0.447  | 0.488 |
| 10% removed        | 0.386 | 0.404 | 0.367 | 0.444  | 0.468 |
| 30% removed        | 0.398 | 0.393 | 0.358 | 0.412  | 0.439 |
| 50% removed        | 0.346 | 0.415 | 0.345 | 0.395  | 0.424 |
| 10% spline         | 0.376 | 0.405 | 0.404 | —      | —     |
| 30% spline         | 0.386 | 0.399 | 0.393 | 0.453  | 0.484 |
| 50% spline         | 0.396 | 0.417 | 0.406 | 0.434  | 0.472 |
| Noise 5 dB         | 0.342 | 0.420 | 0.405 | 0.462  | 0.484 |
| Jitter $\pm 50$ ms | 0.389 | 0.418 | 0.421 | 0.447  | 0.472 |
| 1 trial            | 0.367 | 0.367 | 0.396 | 0.363  | 0.358 |
| 4 trials           | 0.349 | 0.406 | 0.377 | 0.392  | 0.411 |
| 16 trials          | 0.368 | 0.399 | 0.393 | 0.466  | 0.472 |

comparison that zero-fills is therefore a comparison of coordinate systems as much as of representations, and its outcome scales with how much each encoder has to lose. This agrees with the benchmark finding that channel-dropout fragility of EEG foundation models depends on whether dropped channels are removed or zero-padded [22], and Proposition 1 with the origin sweep gives the mechanism and its size for standardized features. The reservoir’s indifference to the fill belongs to its PCA coordinates: its raw spike counts are as far from centered as band-power, and moved far from its own mean its code degrades too.

*The empty baseline is not empty after normalization.* Per-epoch z-scoring is a common default, and Proposition 2 shows that it writes  $-\frac{|Q|}{T} m_Q/s$  into every pre-stimulus sample. On the long epoch that is enough for samples recorded before the picture to classify affect above chance, and the same normalization removes amplitude information from the post-stimulus window. The mechanism resembles the way high-pass filters move late ERP effects to earlier latencies [38]–

[40], and, because the normalization uses the whole epoch, it is a form of leakage from the post-stimulus signal into the baseline [41]. For the conference reservoir, whose window lies mostly before onset, the SHAPE accuracy cannot be credited to evoked dynamics inside that window. On the external cohort a window placed after onset turned a chance-level reservoir into one whose point estimate exceeds the ERP-window encoder’s, which is the measurement the operator was meant to make.

*The edge of stability.* The conference reported a transition near  $\rho^* \approx 0.82$  for one reservoir draw. Ten draws on a second cohort’s drives place it at  $0.76 \pm 0.04$ , always below  $\rho = 1$ , but the firing threshold moves it from 0.35 to at least 1.57. For a thresholded spiking reservoir the edge is therefore a measured property of the unit and its drive rather than a value fixed by the spectral radius, in line with edge-of-chaos analyses of other reservoirs [5], [6], [34] and with the result that  $\rho < 1$  is neither necessary nor sufficient for the echo-state property [35]. Clean accuracy was broad around the transition on both cohorts. Its sharpest dependence on the operating point appeared under electrode removal: past the transition a silent electrode lies far from the reservoir’s typical code, and on the external long epoch accuracy with 30% of the electrodes silent fell close to chance on average, although SHAPE’s short window showed no such separation. An operating point chosen near the edge for clean accuracy may thus be the one least tolerant of absent inputs, and measuring both is cheap.

*Training and physical repair.* Two conference conclusions about electrode loss change with the second cohort. First, augmentation did not reduce EEGNet’s loss to electrode removal on the external cohort, so the conference finding that the training procedure sets this architecture’s accuracy under electrode loss holds for SHAPE and the recipe used there, not in general (Table IX varies the recipe). Second, physical repair changed the ranking that the zero trace produced. With spline-repaired electrodes no fixed encoder lost more than 0.011 BA at 50% removal on the long epoch, EEGNet recovered most of its removal loss, and the ERP-window lead over the reservoir disappeared, although at the conference-matched epoch, where the encoders are near chance, the lead persisted after repair. A

zero trace is itself a fill, the signal-level counterpart of zero-fill, and the same advice applies: state it, and compare it with a physically motivated repair [25] before ranking representations by their tolerance of missing electrodes.

*A reporting standard.* The results support five reporting items for EEG representation studies under perturbation. First, absolute accuracy with subject-level uncertainty and an empirical permutation floor [36], [37], never retention ratios alone. Second, the fill rule, varied, together with the mean  $|\mu|/\sigma$  of the features and the above-chance margin, since together they bound how much the rule can matter. Third, the normalization, and whether any window used for decoding precedes onset. Fourth, trained-model results with and without augmentation [24]. Fifth, for reservoirs, the measured transition  $\rho^*$  and more than one weight draw.

*Limitations.* SHAPE could not be reanalyzed for this version, so the new experiments ran on the external cohort only, whose above-chance margin is small. Effects that need headroom were therefore measurable on SHAPE but not on the external cohort, and only their mechanism could be tested on both. The external analysis used one pre-specified long epoch beyond the conference-matched one, and no other epochs were analyzed. The reservoir’s draws, thresholds, and leaks were varied only around the conference design, EEGNet is the only trained architecture, and DEAP was not reanalyzed. The dataset authors interpolated some bad channels before release, which may favor spline repair on that cohort. Two analyses would close these gaps: rerunning SHAPE with a post-onset reservoir window and a fixed-scale normalization, which tests Proposition 2 where the condition effects are large, and repeating the origin sweep on a cohort with SHAPE-sized margins. Clinical labels, graph topology, and structure–function coupling are outside the scope of this paper.

## VII. CONCLUSION

What an evaluation lets stand for nothing is part of the measurement. A zero-filled electrode sits at the coordinate origin, so its effect on dropout accuracy is set by how far that origin lies from the data and by how much the encoder has above chance. Per-epoch normalization writes the evoked signal into the baseline, so the empty interval before onset can carry condition information while the evoked signal itself loses some. The edge of stability of a fixed spiking reservoir is a measured quantity that moves with the weight draw and, more strongly, with the firing threshold, so the echo-state heuristic  $\rho = 1$  marks no transition for this unit, and silence lies farther from the reservoir’s typical state the more active the reservoir is. On two cohorts the reservoir is not the most accurate encoder, but its near-centered code and its controllable dynamics make each of these effects visible, and with its window placed after onset it measures the evoked signal it was built to measure.

## ETHICS

The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory’s Institutional Review Board oversight.

The external cohort and DEAP are publicly released datasets governed by their original collection protocols [28], [32], [33].

## DATA AVAILABILITY

For the SHAPE dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [29]. The external cohort is public under a CC BY-NC 4.0 licence [33], and DEAP is public [28]. The analysis code, which produces every number, table, and figure from these data, and the aggregate results are available from the authors.

## APPENDIX A SUPPLEMENTARY TABLES

Table VII gives the signal-level results of the external cohort at the conference-matched epoch, where every fixed encoder is near chance, and Table VIII the origin test at that epoch. Table IX reports EEGNet across training recipes on the seed-42 partition of the conference-matched epoch.

TABLE VII  
EXTERNAL COHORT, CONFERENCE-MATCHED EPOCH (−200 TO +800 MS): SUBJECT-LEVEL BA UNDER THE SIGNAL-LEVEL CONDITIONS OF TABLE VI.

| Condition          | Band  | ERP   | Res.  | EEGNet | +aug. |
|--------------------|-------|-------|-------|--------|-------|
| Clean              | 0.368 | 0.386 | 0.320 | 0.501  | 0.523 |
| 10% removed        | 0.352 | 0.379 | 0.300 | 0.465  | 0.497 |
| 30% removed        | 0.360 | 0.370 | 0.310 | 0.443  | 0.472 |
| 50% removed        | 0.342 | 0.349 | 0.301 | 0.452  | 0.466 |
| 10% spline         | 0.367 | 0.392 | 0.326 | –      | –     |
| 30% spline         | 0.361 | 0.374 | 0.351 | 0.487  | 0.516 |
| 50% spline         | 0.357 | 0.401 | 0.346 | 0.493  | 0.497 |
| Noise 5 dB         | 0.354 | 0.392 | 0.370 | 0.459  | 0.500 |
| Jitter $\pm 50$ ms | 0.365 | 0.404 | 0.355 | 0.477  | 0.480 |
| 1 trial            | 0.329 | 0.330 | 0.357 | 0.370  | 0.377 |
| 4 trials           | 0.358 | 0.383 | 0.344 | 0.424  | 0.431 |
| 16 trials          | 0.361 | 0.392 | 0.346 | 0.484  | 0.482 |

TABLE VIII  
ORIGIN TEST AT 30% DROPOUT, EXTERNAL COHORT, CONFERENCE-MATCHED EPOCH (COLUMNS AS IN TABLE III).

| Encoder    | $\frac{ \mu }{\sigma}$ | Clean | Zero  | $\kappa=-2$ | $\kappa=0$ | $\kappa=+2$ | $\ \Delta\ell\ $ | Flip % |
|------------|------------------------|-------|-------|-------------|------------|-------------|------------------|--------|
| Band-power | 1.20                   | 0.368 | 0.360 | 0.342       | 0.360      | 0.341       | 1.78             | 37.8   |
| ERP-window | 0.67                   | 0.386 | 0.370 | 0.351       | 0.354      | 0.346       | 0.71             | 25.9   |
| Reservoir  | 0.05                   | 0.320 | 0.325 | 0.303       | 0.325      | 0.325       | 0.17             | 3.3    |

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TABLE IX

EEGNET ACROSS TRAINING RECIPES (EXTERNAL COHORT, CONFERENCE-MATCHED EPOCH, SEED-42 PARTITION): SUBJECT-LEVEL BA CLEAN, WITH 30% AND 50% OF ELECTRODES REMOVED, AND UNDER 5 dB NOISE, WITHOUT AND WITH AUGMENTATION, AND THE PAIRED AUGMENTATION EFFECT AT 50% REMOVAL.

| Recipe                      | EEGNet |       |       |       | EEGNet + aug. |       |       |       | Aug.—unaug., 50%<br>[95% CI] |
|-----------------------------|--------|-------|-------|-------|---------------|-------|-------|-------|------------------------------|
|                             | Clean  | 30%   | 50%   | Noise | Clean         | 30%   | 50%   | Noise |                              |
| Default (80 ep., $p=0.25$ ) | 0.463  | 0.408 | 0.399 | 0.434 | 0.485         | 0.447 | 0.446 | 0.465 | +0.047 [+0.013, +0.079]      |
| 40 epochs                   | 0.459  | 0.417 | 0.404 | 0.430 | 0.472         | 0.442 | 0.424 | 0.468 | +0.020 [−0.009, +0.048]      |

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